

Danmarks
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02417 - TIME SERIES ANALYSIS

Assignment 2

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1 Problem 1

1.1

We have the process

$$X_t + \phi_1 X_{t-1} + \phi_2 X_{t-2} = 0 \quad (1.1)$$

We write the characteristic polynomial for X_t , with

$$\phi_1 = -0.7, \quad \phi_2 = -0.2 \quad (1.2)$$

Starting from the model in Equation 1.1, the characteristic polynomial becomes

$$\phi_2 z^2 + \phi_1 z + 1 = 0 \quad (1.3)$$

Substituting the parameter values from Equation 1.2 gives

$$-0.2z^2 - 0.7z + 1 = 0 \quad (1.4)$$

We solve for the roots

$$z_1 = -4.59, \quad z_2 = 1.09 \quad (1.5)$$

Next we compute the inverse of the roots

$$\frac{1}{z_1} = -0.218, \quad \frac{1}{z_2} = 0.917 \quad (1.6)$$

Since both

$$\left| \frac{1}{z_1} \right| < 1 \quad (1.7)$$

and

$$\left| \frac{1}{z_2} \right| < 1 \quad (1.8)$$

the roots lie within the unit circle, and from slide 13/36 in lecture 5 it must therefore hold that the process is stationary.

1.2

A known property of $AR(p)$ models is that they are always invertible (slide 13/36 lecture 5). Hence, Process 1.1 is invertible.

1.3

We wish to find an expression for the autocorrelation $\rho(k)$ for Equation 1.1 as a function of ϕ_1 and ϕ_2 . By using the statement for the autocorrelation function on slide 3/36 in lecture 6 together with Eq. 5.81 in the textbook, the autocorrelation function satisfies

$$\rho(k) = -\phi_1 \rho(k-1) - \phi_2 \rho(k-2), \quad k \geq 2. \quad (1.9)$$

In other words, the new autocorrelation is found from the two previous autocorrelations. We therefore determine $\rho(0)$ and $\rho(1)$, which serve as initial values for the recursion. By definition, a variable has correlation 1 with itself, hence

$$\rho(0) = 1. \quad (1.10)$$

We now find $\rho(1)$ from Eq. 5.81 with $p = 2$:

$$\rho(1) = -\phi_1 - \phi_2 \rho(1) \quad (1.11)$$

$$\rho(1) + \phi_2 \rho(1) = -\phi_1 \quad (1.12)$$

$$(1 + \phi_2)\rho(1) = -\phi_1 \quad (1.13)$$

$$\rho(1) = \frac{-\phi_1}{1 + \phi_2}. \quad (1.14)$$

1.4

We plot the autocorrelation function 1.9 up to lag 30 with the provided values 1.2.

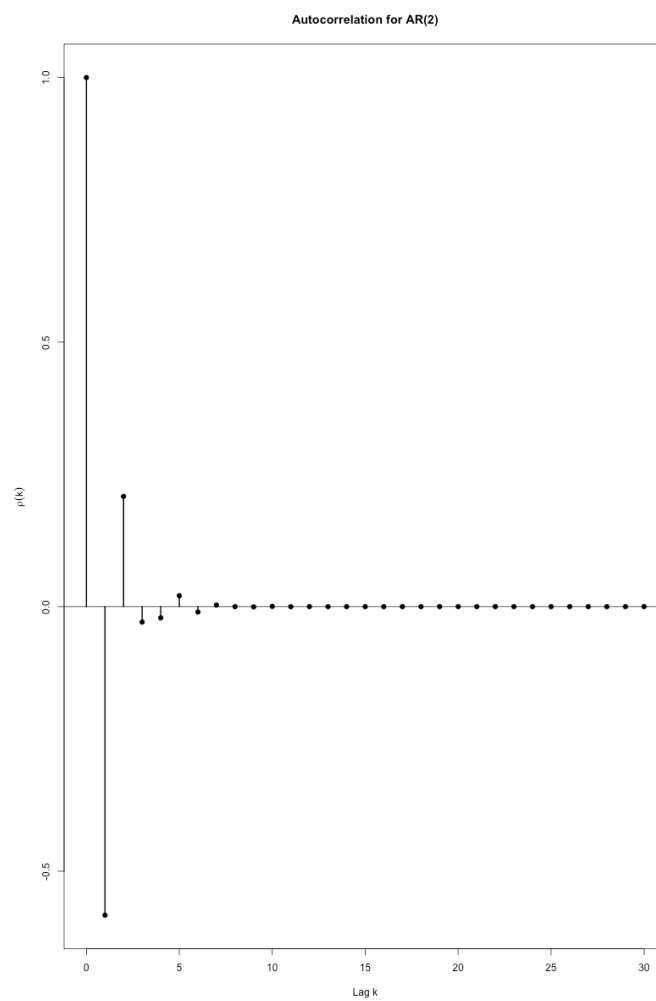


Figure 1: ACF plot of process 1.1 to lag 30.

As expected from the golden table for *ARMA* identification, the ACF displays damped sinusoidal oscillations, which are indicative of an *AR*(*p*) process.

2 Problem 2 Simulating Seasonal Processes

A multiplicative seasonal ARIMA model $(p,d,q) \times (P,D,Q)_s$ is given by

$$\phi(B)\Phi(B^s)\nabla^d\nabla_s^D Y_t = \theta(B)\Theta(B^s)\varepsilon_t,$$

where $\{\varepsilon_t\}$ is white noise. All models below have $d = D = 0$ and $s = 12$. The multiplicative structure means the non-seasonal and seasonal polynomials interact, producing ACF/PACF spikes at both short lags and multiples of $s = 12$.

2.1 $(1,0,0) \times (0,0,0)_{12}$, $\phi_1 = 0.6$

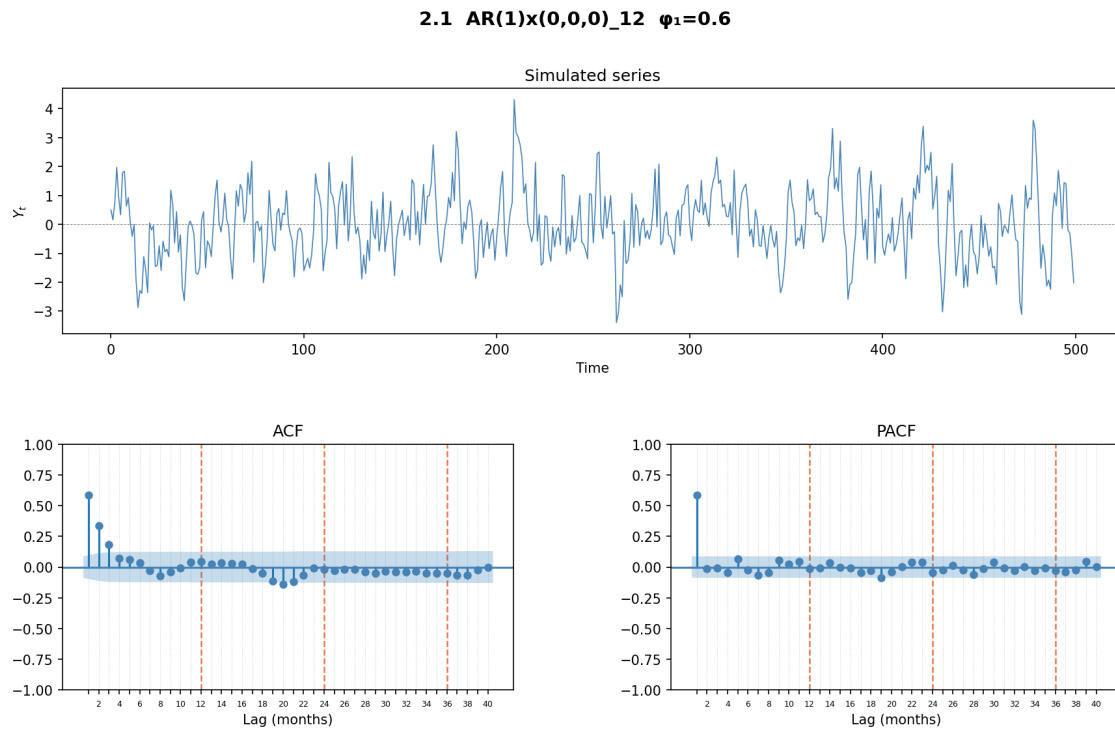


Figure 2: $(1,0,0) \times (0,0,0)_{12}$, $\phi_1 = 0.6$.

A plain AR(1) without seasonal structure: $Y_t = 0.6Y_{t-1} + \varepsilon_t$. The ACF decays exponentially and the PACF cuts off sharply after lag 1.

2.2 $(0,0,0) \times (1,0,0)_{12}$, $\Phi_1 = -0.9$

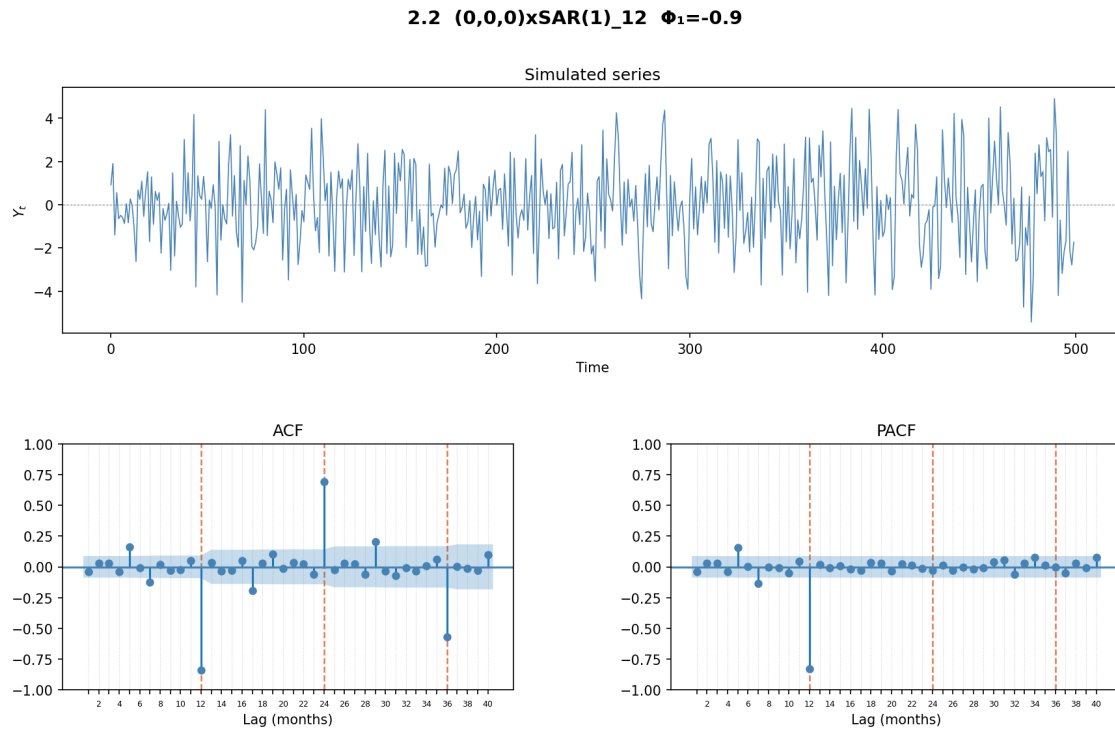


Figure 3: $(0,0,0) \times (1,0,0)_{12}$, $\Phi_1 = -0.9$.

A seasonal AR(1): $Y_t = -0.9Y_{t-12} + \varepsilon_t$. The ACF spikes at lags 12, 24, 36, ... with alternating signs (due to $\Phi_1 < 0$), and the PACF cuts off after lag 12.

2.3 $(1,0,0) \times (0,0,1)_{12}$, $\phi_1 = 0.9$, $\Theta_1 = -0.7$

2.3 AR(1)xSMA(1)_12 $\phi_1=0.9$, $\Theta_1=-0.7$

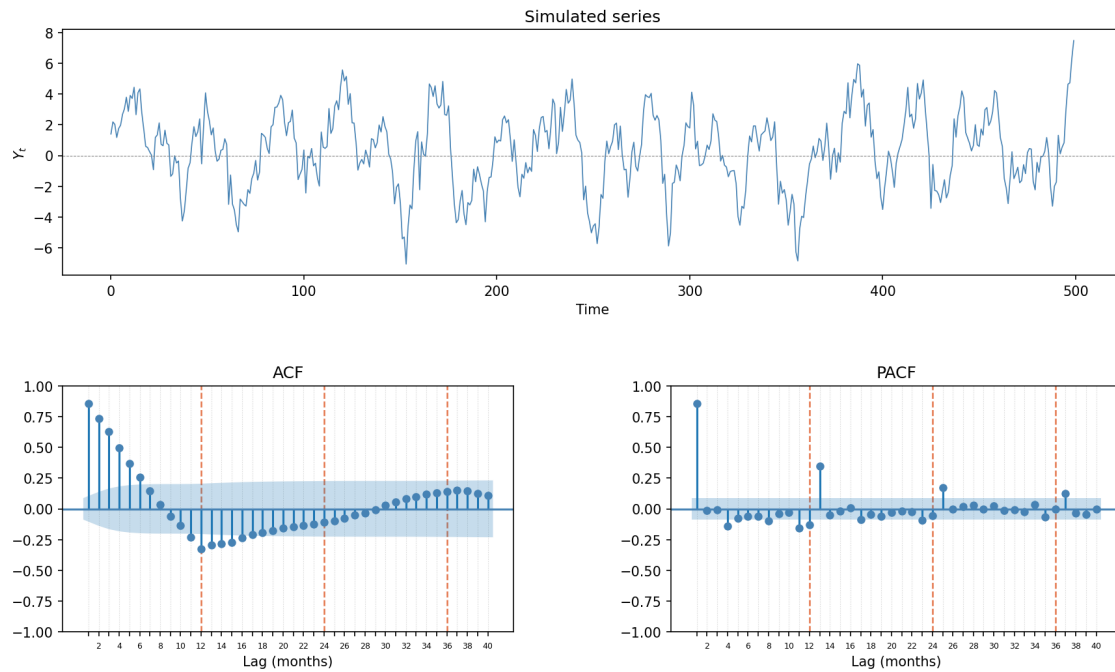


Figure 4: $(1,0,0) \times (0,0,1)_{12}$, $\phi_1 = 0.9$, $\Theta_1 = -0.7$.

The high ϕ_1 creates a slow oscillating decay at in the ACF. The PACF tails off at seasonal lags.

2.4 $(1,0,0) \times (1,0,0)_{12}$, $\phi_1 = -0.6$, $\Phi_1 = -0.8$

2.4 AR(1)xSAR(1)_12 $\phi_1=-0.6$, $\Phi_1=-0.8$

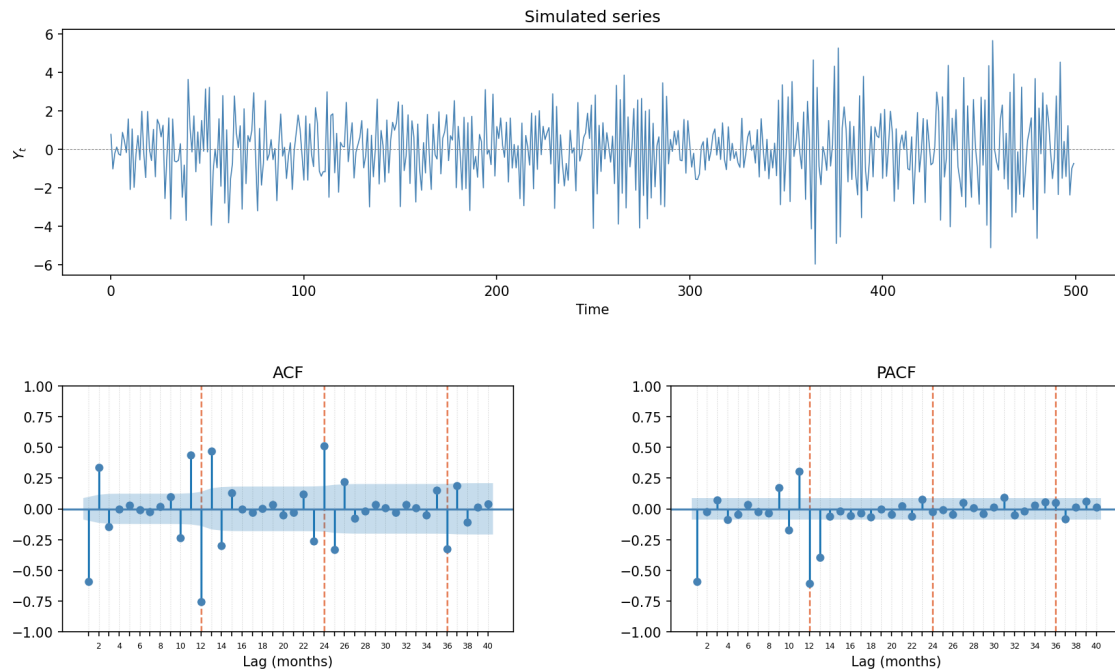


Figure 5: $(1,0,0) \times (1,0,0)_{12}$, $\phi_1 = -0.6$, $\Phi_1 = -0.8$.

The ACF tails off at both short lags (alternating due to $\phi_1 < 0$) and seasonal lags (alternating due to $\Phi_1 < 0$). The PACF has spikes at lags 1, 12 and 13, then cuts off.

2.5 $(0,0,1) \times (0,0,1)_{12}$, $\theta_1 = 0.4$, $\Theta_1 = -0.8$

2.5 MA(1)xSMA(1)_12 $\theta_1=0.4$, $\Theta_1=-0.8$

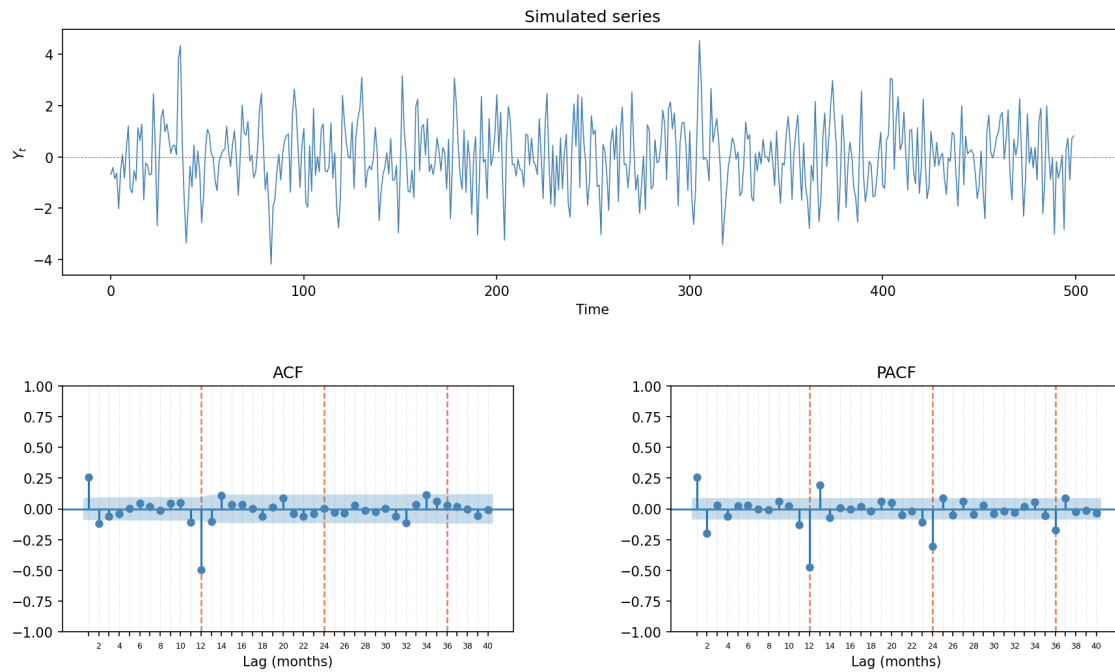


Figure 6: $(0,0,1) \times (0,0,1)_{12}$, $\theta_1 = 0.4$, $\Theta_1 = -0.8$.

The ACF cuts off after lag 1 and shows spikes at lag 12 only, then is statistically zero. The PACF tails off slowly at seasonal lags, reflecting the underlying MA structure.

2.6 $(0,0,1) \times (1,0,0)_{12}$, $\theta_1 = -0.4$, $\Phi_1 = 0.7$

2.6 MA(1)xSAR(1)_12 $\theta_1=-0.4$, $\Phi_1=0.7$

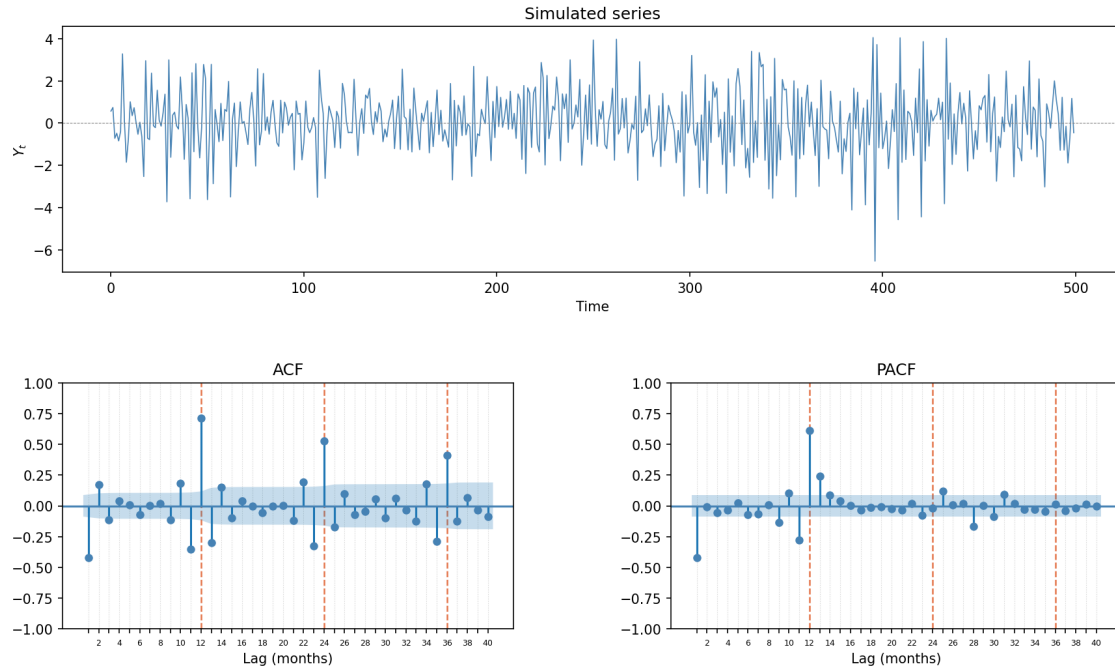


Figure 7: $(0,0,1) \times (1,0,0)_{12}$, $\theta_1 = -0.4$, $\Phi_1 = 0.7$.

The MA term at lag 1 creates a single spike in the ACF at lag 1, while the seasonal AR term causes the ACF to tail off at lags 12, 24, 36. The PACF has a spike at lag 1 and tails off at seasonal lags.

2.7 Summary

The simulations reveal that the standard ARMA identification rules extend directly to the seasonal frequency s . A seasonal $AR(P)_s$ component causes the ACF to tail off at lags $s, 2s, 3s, \dots$ with the PACF cutting off after lag Ps , while a seasonal $MA(Q)_s$ component reverses this: the ACF cuts off and the PACF tails off at seasonal lags. The multiplicative structure introduces cross-terms at lags $s \pm p$ (or $s \pm q$) with magnitude approximately $\phi_1 \theta_1$, and negative coefficients cause alternating signs in the ACF at the relevant lags. Crucially, the short-lag and seasonal-lag behaviour can largely be read off independently, making identification feasible by examining the two frequency ranges separately.

3 Problem 3

In this section, we guess the ARMA model structure of each process based on the time series plot together with its sample ACF and PACF. The identification is based on the standard rules: for an AR process, the ACF typically tails off while the PACF cuts off after a small number of lags; for an MA process, the ACF cuts off while the PACF tails off; and for an ARMA process, both the ACF and PACF usually tail off gradually.

3.1 Process 1

Guess: ARMA(0,0), i.e. white noise.

The sample ACF is close to zero for almost all nonzero lags, with only a few small random spikes. The PACF also shows no clear pattern and does not cut off after any particular lag. This suggests that there is no meaningful serial dependence in the series, so the process is best identified as white noise.

3.2 Process 2

Guess: AR(2).

For Process 2, the sample ACF decays gradually rather than cutting off sharply, which is typical of an autoregressive process. At the same time, the PACF has two clearly significant spikes at lags 1 and 2, after which the remaining values are relatively small. This is the standard pattern of an AR(2) process, so AR(2) is the most reasonable choice.

3.3 Process 3

Guess: ARMA(1,2).

In this case, neither the ACF nor the PACF cuts off sharply. Instead, both functions decay gradually, which suggests an ARMA model rather than a pure AR or MA model. A reasonable low-order choice is ARMA(1,2), since the ACF shows a strong positive dependence at small lags that slowly dies out, while the PACF also has several important early spikes without a clean cutoff. Therefore, Process 3 is best identified as an ARMA(1,2) process.

3.4 Conclusion

Our final guesses for the three processes are:

- Process 1: ARMA(0,0) (white noise),
- Process 2: AR(2),
- Process 3: ARMA(1,2).

4 Use of AI

4.1 Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Github Copilot and chatGPT in order to

- develop/debug code,
- improve grammar,
- make corrections,
- assist with LaTeX formatting.

After using this tool/service, the authors reviewed and edited the content as needed and take(s) full responsibility for the content of the publication